

```
#include <iostream> using namespace std;
```

```
void countingSort(int arr[], int n) {  
    int max = arr[0];  
    for (int i = 1; i < n; i++) {  
        if (arr[i] > max) {  
            max = arr[i];  
        }  
    }  
  
    int count[100] = {0};  
    for (int i = 0; i < n; i++) {  
        count[arr[i]]++;  
    }  
  
    for (int i = 1; i <= max; i++) {  
        count[i] += count[i - 1];  
    }  
  
    int output[100];  
    for (int i = n - 1; i >= 0; i--) {  
        output[count[arr[i]] - 1] = arr[i];  
        count[arr[i]]--;  
    }  
  
    for (int i = 0; i < n; i++) {  
        arr[i] = output[i];  
    }  
}
```

```
int main() {  
    int arr[] = {4, 2, 2, 8, 3, 3, 1};  
    int n = sizeof(arr) / sizeof(arr[0]);  
  
    cout << "Original array: ";  
    for (int i = 0; i < n; i++) {  
        cout << arr[i] << " ";  
    }  
    cout << endl;  
  
    countingSort(arr, n);  
  
    cout << "Sorted array: ";  
    for (int i = 0; i < n; i++) {  
        cout << arr[i] << " ";  
    }  
    cout << endl;  
  
    return 0; }
```